

2. Compounds these returns to show the growth of our investment over time
3. Compares our strategy's performance to a simple buy-and-hold approach
4. Calculates some basic performance metrics

While this approach gives us a first look at our strategy's performance, it's quite simplistic. It doesn't account for important real-world factors like:

- Trading costs (commissions, slippage)
- Market impact (your trades affecting the price)
- Realistic trade execution (getting the exact prices shown in the data)
- Risk management (position sizing, stop losses)

For more realistic backtesting, we need specialized frameworks.

Backtesting Frameworks

While our simple backtest gives us a basic understanding of the strategy's performance, professional algorithmic traders use specialized frameworks for more realistic and comprehensive testing. Let's explore two popular Python backtesting frameworks.

Backtrader: A Comprehensive Python Framework

Backtrader is a popular, feature-rich Python framework for backtesting trading strategies. It offers realistic order execution, multiple data feeds, and sophisticated analysis tools.

Here's a simplified example of how we might implement our moving average crossover strategy in Backtrader:

```
1. import backtrader as bt
2.
```